Homework 3

1. Introduction

This report presents an implementation of image-based malware family classification, following the method introduced by Nataraj et al. (2011). Malware binaries are converted into grayscale images by interpreting their raw bytes as pixel intensities, exploiting the observation that samples from the same family exhibit visually similar structural and textural patterns when rendered this way. Using 485 disarmed samples spanning 8 malware families from the MOTIF dataset, three classification approaches were implemented and compared: a Support Vector Machine (SVM) trained on Histogram of Oriented Gradients (HOG) features, an SVM trained on Local Binary Pattern (LBP) features, and a Convolutional Neural Network (CNN) trained end-to-end on the raw grayscale images. The CNN achieved the highest test accuracy at 75.3%, followed by HOG+SVM at 71.1% and LBP+SVM at 58.8%, indicating that both learned convolutional features and hand-crafted gradient-based descriptors substantially outperform purely local texture statistics for this task.

2. Dataset

Samples were drawn from the MOTIF (Malware Open-source Threat Intelligence Family) dataset (Joyce et al., 2022), which provides 3,095 disarmed Windows PE malware samples across 454 families, each labeled with a ground-truth family name derived from published threat intelligence reports. Disarmed samples are intentionally modified by the dataset authors so they cannot execute, while otherwise preserving the byte-level structure relevant to static analysis. Per the assignment's requirements, no pre-transformed dataset (such as MalImg) was used; all image conversion was performed directly from the raw MOTIF binaries using custom code.

From the 454 available families, the 8 families with the largest sample counts were selected to ensure sufficient data per class for training and evaluation:

Family	Samples
icedid	142
azorult	68

phoripiex	58
maze	52
trickbot	43
gandcrab	41
locky	41
seduploader	40

The selected families exhibit moderate class imbalance (40–142 samples per family), which is representative of real-world malware prevalence and was accounted for during evaluation via stratified train/test splitting and per-class precision/recall reporting rather than relying on overall accuracy alone.

All handling of malware samples was performed in an isolated virtual machine (VirtualBox, Ubuntu) with network access disabled except during dataset download. Samples were only ever opened in binary read mode for byte extraction; at no point were any samples executed.

3. Methodology

The pipeline consists of four stages: (1) sample organization by family, (2) binary-to-image conversion, (3) feature extraction, and (4) model training and evaluation.

a. Sample Organization

Sample metadata (MD5 hash and reported family name) was parsed from the MOTIF dataset's motif_dataset.jsonl file. Family sample counts were tallied, the top 8 families by count were selected, and the corresponding binaries were copied from the raw sample pool into per-family directories to prepare for image conversion.

b. Binary-to-image Conversion

Each binary was read as a sequence of unsigned 8-bit integers (byte values 0–255) and reshaped into a 2D array, interpreted directly as pixel intensities of a grayscale image (0 = black, 255 = white). Image width was selected according to file size, following the empirically-derived width table from the assignment specification (e.g. files under 10 KB use width 32; files between 100–200 KB use width 384; files over 1000 KB use width 1024). Image height was determined by the number of complete rows of that width the byte stream could fill; trailing bytes that did not complete a full row were discarded. This produced one grayscale PNG image per binary, preserving the family-based directory structure

c. Feature Extraction

All images were resized to a fixed 128×128 resolution to produce feature vectors of consistent length across samples, since raw image heights vary with original file size. Two texture-based descriptors were extracted:

- Histogram of Oriented Gradients (HOG): captures local edge and gradient orientation structure, producing an 8,100-dimensional feature vector per image (9 orientation bins, 8×8 pixel cells, 2×2 cell blocks with L2-Hys normalization).
- Local Binary Patterns (LBP): captures local micro-texture by thresholding each pixel's neighborhood, producing a 10-dimensional uniform-pattern histogram per image (P=8 neighbors, R=1 radius).

d. Classification Models

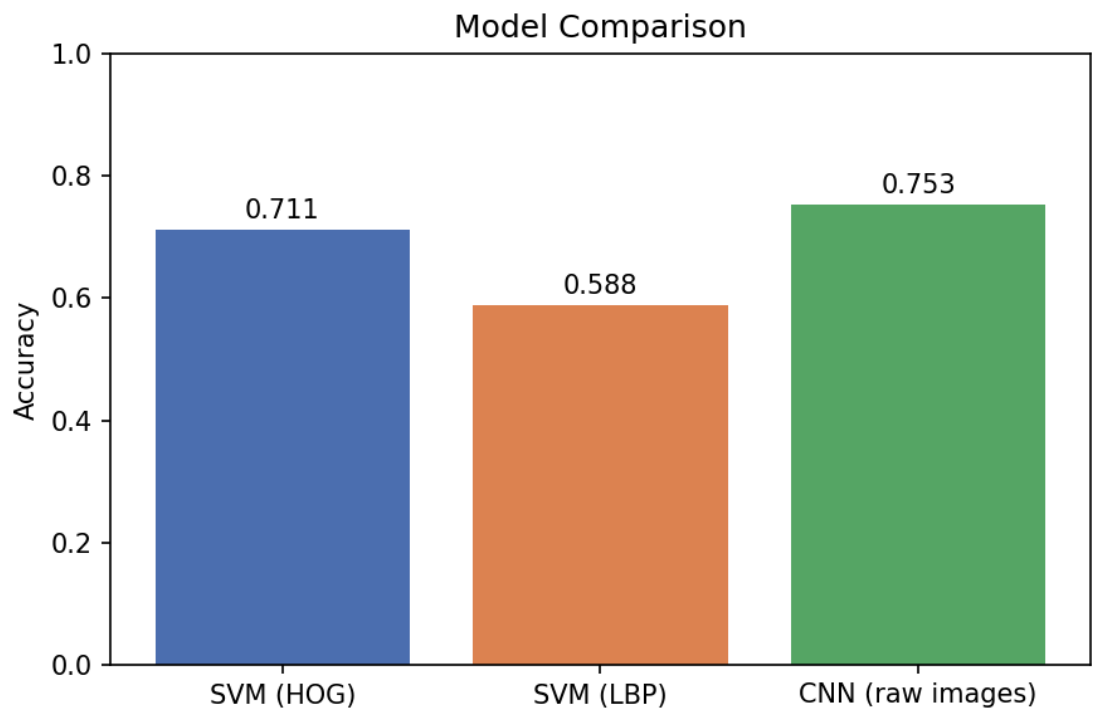
Three classifier configurations were trained and compared:

1. SVM (HOG features) — RBF-kernel Support Vector Machine (C=10) trained on standardized HOG feature vectors.
2. SVM (LBP features) — RBF-kernel SVM trained on standardized LBP histograms.
3. CNN (raw images) — A compact convolutional neural network trained end-to-end on raw 128×128 grayscale images, consisting of three convolution + max-pooling blocks (16 → 32 → 64 channels) followed by a fully connected classification head with dropout regularization. Trained for 15 epochs using the Adam optimizer and cross-entropy loss.

For all three models, data was split into training (80%) and test (20%) sets using stratified sampling to preserve family proportions across splits, given the moderate class imbalance in the dataset.

4. Results

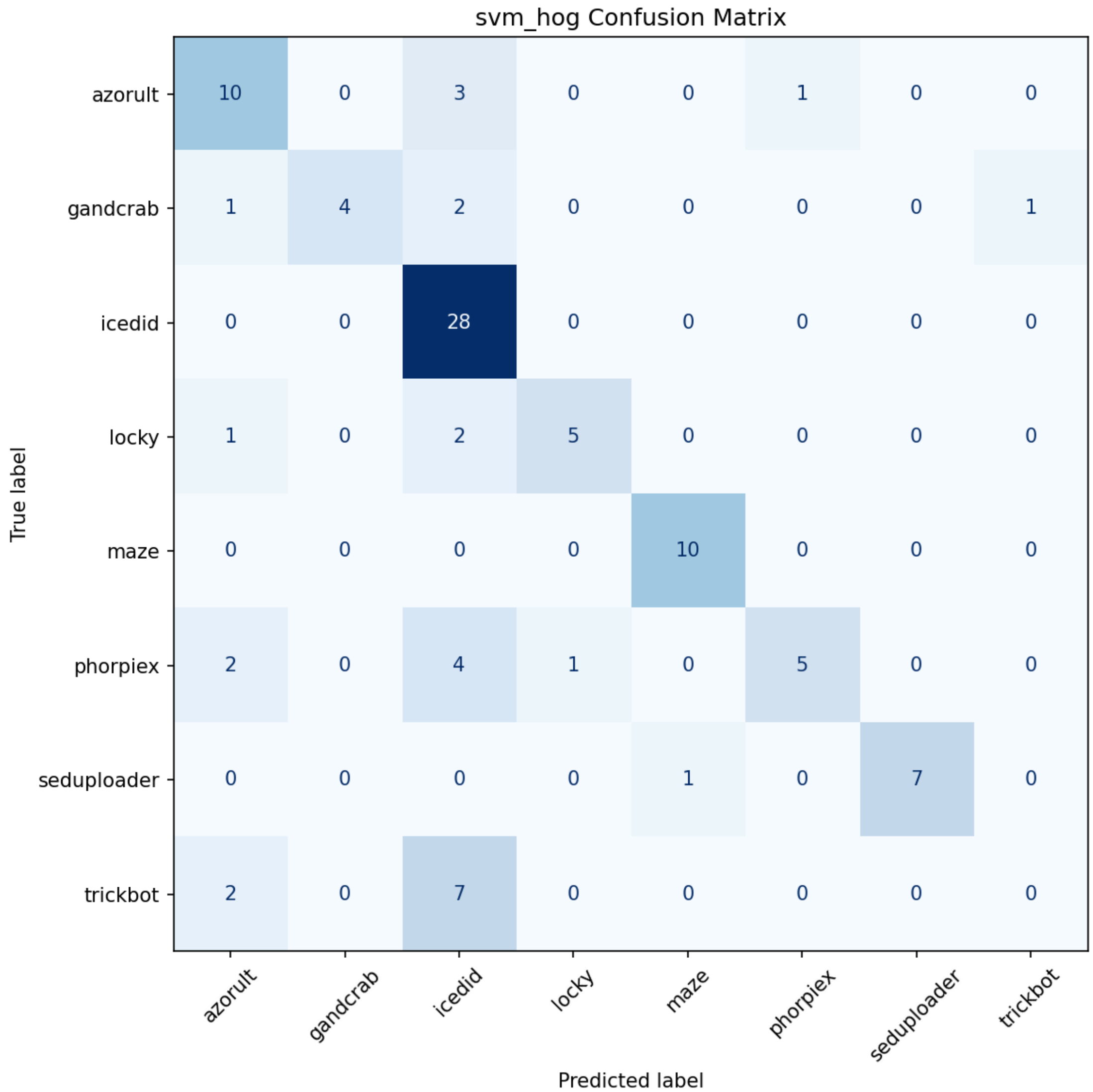
Method	Test Accuracy
SVM (HOG Features)	71.1%
SVM (LBP Features)	58.8%
CNN (raw images)	75.3%



a. HOG + SVM

Family	Precision	Recall	F1-Score	Support
azorult	0.62	0.71	0.67	14
gandcrab	1.00	0.5	0.67	8
icedid	0.61	1.00	0.76	28
locky	0.83	0.62	0.91	8
maze	0.91	1.00	0.95	10
phorpiex	0.83	0.42	0.56	12

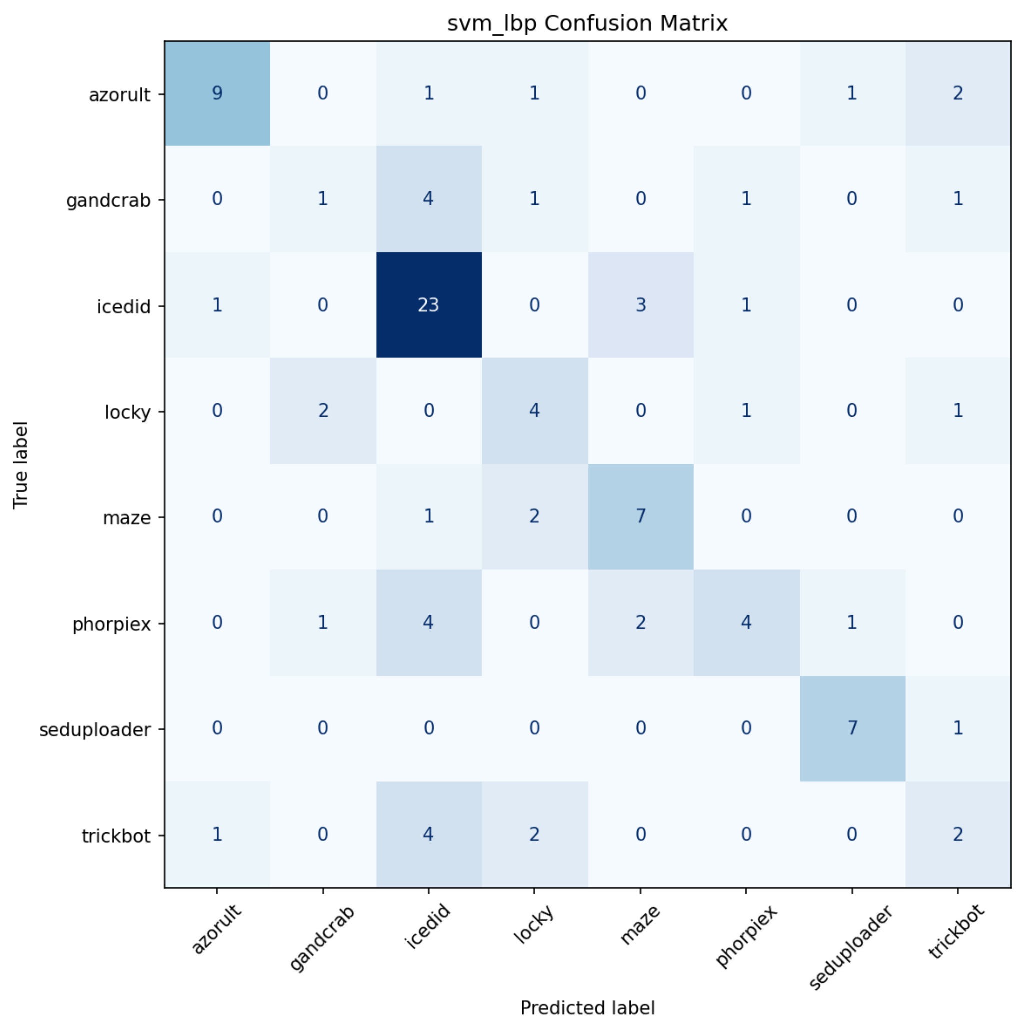
seduploader	2.00	0.88	0.93	8
trickbot	0.00	0.00	0.00	9



b. LBP + SVM

Family	Precision	Recall	F1-Score	Support
azorult	0.80	0.64	0.72	14
gandcrab	0.25	0.12	0.17	8

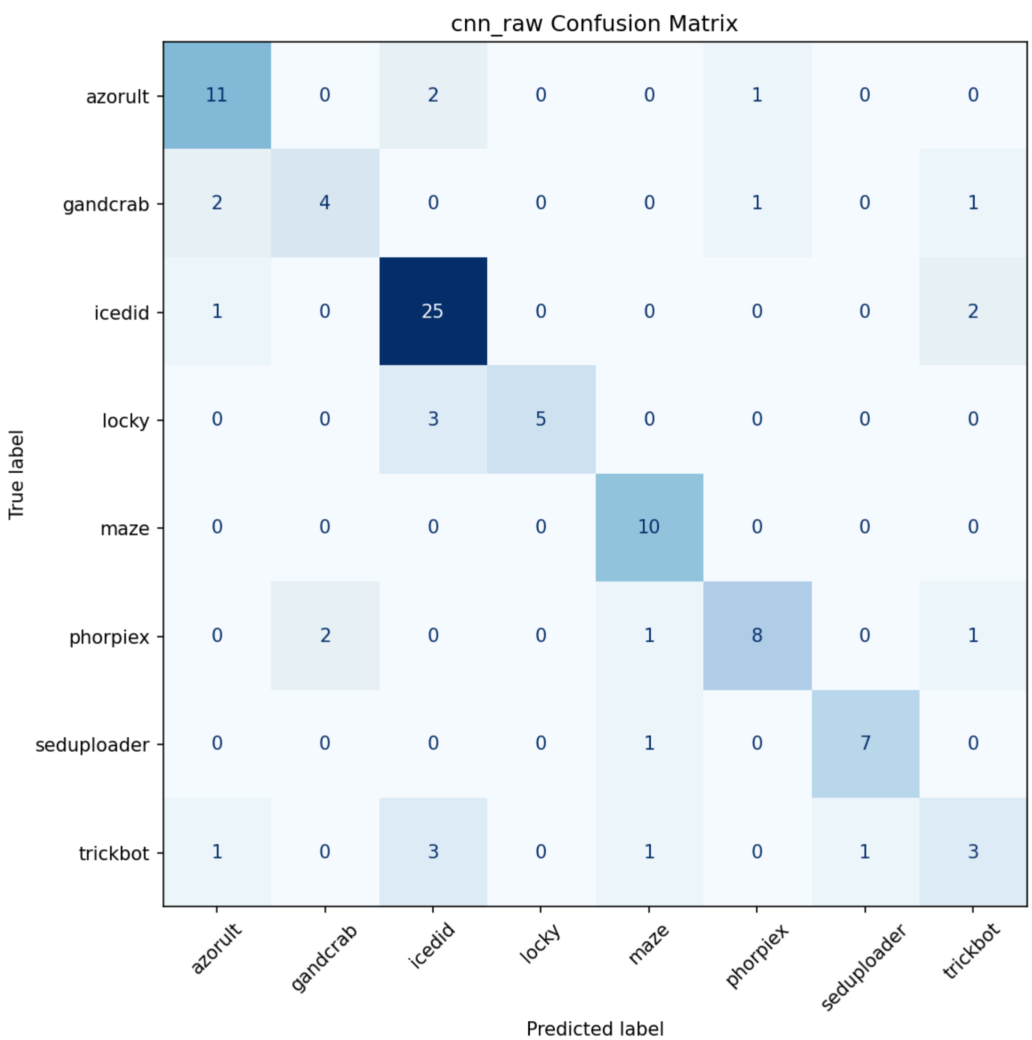
icedid	0.62	0.82	0.71	28
locky	0.40	0.50	0.44	8
maze	0.58	0.70	0.64	10
phorpiex	0.57	0.33	0.42	12
seduploader	0.78	0.88	0.82	8
trickbot	0.29	0.22	0.25	9



c. CNN

Family	Precision	Recall	F1-Score	Support

azorult	0.73	0.79	0.76	14
gandcrab	0.67	0.50	0.57	8
icedid	0.76	0.89	0.82	28
locky	1.00	0.62	0.77	8
maze	0.77	1.00	0.87	10
phorpiex	0.80	0.67	0.73	12
seduploader	0.88	0.88	0.88	8
trickbot	0.43	0.33	0.38	9



d. Discussion

The CNN trained directly on raw grayscale images achieved the best overall accuracy (75.3%), outperforming both hand-crafted feature approaches. This is consistent with expectations: convolutional layers can learn hierarchical, task-specific visual features rather than relying on fixed, general-purpose descriptors, and given a training set of roughly 388 images (80% of 485) the model had enough data to learn meaningful low-level texture filters without severely overfitting, as reflected in the steadily decreasing training loss (1.99 → 0.32 over 15 epochs) without signs of instability.

Between the two hand-crafted descriptors, HOG substantially outperformed LBP (71.1% vs. 58.8%). This suggests that gradient orientation and edge structure are more discriminative for these malware images than the fine-grained local intensity patterns captured by LBP. This is plausible given that malware images generated by this method often contain structured regions (e.g. code sections, padding, resource blocks) with directional patterns, which LBP's purely local neighborhood comparisons are less suited to capture at a global level.

A consistent weak point across all three methods was the trickbot family, which was classified poorly by every model (F1-scores of 0.00, 0.25, and 0.38 for HOG+SVM, LBP+SVM, and CNN respectively). Two likely contributing factors: (1) trickbot had a moderate sample count (43, roughly a third of the largest class), which may not be sufficient for the models to learn its distinguishing visual pattern reliably, and (2) trickbot samples may share structural or packing similarities with other families in the selected set (e.g. azorult or icedid), causing the classifiers to confuse them. [Insert specific misclassification pattern once confusion matrices are reviewed

In contrast, maze and seduploader were classified with high accuracy across all methods (F1 ≥ 0.87 for CNN, ≥ 0.93 for HOG+SVM), suggesting these families produce visually distinctive image textures that are easily separable from the rest of the selected set, regardless of the feature extraction approach used.

The icedid class, being the largest (142 samples, roughly 29% of the dataset), exhibited a precision/recall imbalance in the SVM models — high recall but comparatively lower precision — indicating the classifiers had some tendency to default toward the majority class when uncertain. This is a common effect of class imbalance and could potentially be mitigated in future work with class-weighted loss functions or resampling techniques.

5. Conclusions

This project implemented and evaluated an image-based static malware classification pipeline following Nataraj et al.'s method, using the MOTIF dataset. Converting malware binaries directly into grayscale images proved effective for family classification without requiring execution or disassembly. Among the three approaches compared, a CNN trained directly on raw images achieved the highest accuracy (75.3%), followed by an HOG-based SVM (71.1%) and an LBP-based SVM (58.8%), demonstrating that both learned and hand-crafted gradient-based features are considerably more effective than purely local texture statistics for this task. Future work could explore a larger family set, class-balancing techniques, and additional descriptors such as GIST to further improve classification performance.